weeek6

@relation Iris

@attribute sepallength REAL

@attribute sepalwidth REAL

@attribute petallength REAL

@attribute petalwidth REAL

@attribute class {Iris-setosa,Iris-versicolor,Iris-virginica}

@data

5.1,3.5,1.4,0.2,Iris-setosa

4.9,3.0,1.4,0.2,Iris-setosa

4.7,3.2,1.3,0.2,Iris-setosa

4.6,3.1,1.5,0.2,Iris-setosa

5.0,3.6,1.4,0.2,Iris-setosa

5.4,3.9,1.7,0.4,Iris-setosa

4.6,3.4,1.4,0.3,Iris-setosa

5.0,3.4,1.5,0.2,Iris-setosa

4.4,2.9,1.4,0.2,Iris-setosa

4.9,3.1,1.5,0.1,Iris-setosa

Week1,week7

@relation Student

@attribute age { <30, 30-40, >40 }

@attribute income { low, medium, high }

@attribute student { yes, no }

@attribute credit_rating { fair, excellent }

@attribute marks numeric

@attribute buysPC { yes, no }

@data

<30, high, no, fair, 80, no

<30, high, no, excellent, 40, no

30-40, high, no, fair, 65, yes

>40, medium, no, fair, 90, yes

>40, low, yes, excellent, 99, no

>40, low, yes, fair, 60, yes

30-40, low, yes, excellent, 70, yes

<30, medium, no, fair, 75, no

<30, low, yes, fair, 80, no

>40, medium, yes, fair, 55, yes

>40, medium, yes, excellent, 88, yes

30-40, medium, no, excellent, 78, yes

30-40, high, yes, fair, 45, yes

>40, medium, no, excellent, 40, no

Week2,3

@relation Contact_lenses

@attribute age {young, pre-presbyopic, presbyopic}

@attribute spectacle-prescrip {myope, hypermetrope}

@attribute astigmatism {no, yes}

@attribute tear-prod-rate {reduced, normal}

@attribute contact-lenses {soft, hard, none}

@data

young, myope, no, reduced, none

young, myope, no, normal, soft

young, myope, yes, reduced, none

young, myope, yes, normal, hard

young, hypermetrope, no, reduced, none

young, hypermetrope, no, normal, soft

young, hypermetrope, yes, reduced, none

young, hypermetrope, yes, normal, hard

pre-presbyopic, myope, no, reduced, none

pre-presbyopic, myope, no, normal, soft

pre-presbyopic, myope, yes, reduced, none

pre-presbyopic, myope, yes, normal, hard

pre-presbyopic, hypermetrope, no, reduced, none

pre-presbyopic, hypermetrope, no, normal, soft

pre-presbyopic, hypermetrope, yes, reduced, none

pre-presbyopic, hypermetrope, yes, normal, hard

presbyopic, myope, no, reduced, none

presbyopic, myope, no, normal, none

presbyopic, myope, yes, reduced, none

presbyopic, myope, yes, normal, hard

presbyopic, hypermetrope, no, reduced, none

presbyopic, hypermetrope, no, normal, soft

presbyopic, hypermetrope, yes, reduced, none

presbyopic, hypermetrope, yes, normal, none

week4

@relation Emp

@attribute Age {20,30,40,50}

@attribute Dept {cse,ece,csd,csm}

@attribute Experience {3,4,5,6}

@attribute Salary {50000,60000,70000,80000}

@attribute Attendance {50,60,75,90}

@attribute Performance {good,avg,poor}

@data

20,cse,3,50000,50,poor

30,ece,4,60000,75,avg

40,cse,5,70000,60,good

50,csm,6,80000,90,good

20,csd,4,50000,75,avg

50,cse,6,80000,90,good

20,csm,4,60000,75,avg

50,ece,6,50000,60,avg

40,cse,4,70000,75,good

30,csm,3,50000,90,poor

Week5

@relation Employee

@attribute Age {Young, Middle, Senior}

@attribute Department {IT, HR, Finance, Sales}

@attribute Experience {Low, Medium, High}

@attribute Salary {Low, Medium, High}

@attribute Attendance {Poor, Average, Good}

@attribute Performance {Poor, Average, Good}

@data

Young, IT, Low, Low, Poor, Poor

Middle, HR, Medium, Medium, Average, Average

Senior, Finance, High, High, Good, Good

Young, Sales, Medium, High, Good, Average

Senior, HR, High, High, Average, Good

Middle, Finance, Low, Medium, Poor, Average

Young, HR, Low, Low, Poor, Poor

Middle, Sales, Medium, High, Poor, Good

Senior, IT, High, High, Average, Good

Young, Sales, Medium, Low, Poor, Average

Week 11

1. Normal Distribution –

2. dnorm()

```
x <- seq(-10, 10, by = 0.1)
y <- dnorm(x, mean = 2.0, sd = 1.5)
plot(x, y, main = "Density Normal Distribution",
xlab = "x", ylab = "Density", type = "l")
model <- lm(y ~ x)
abline(model, col = "red", lwd = 2)
```

4. Normal Distribution – pnorm()

```
x <- seq(-10, 10, by = 0.1)
y <- pnorm(x, mean = 2.0, sd = 0.5)
```

```
plot(x, y, main = "Cumulative Normal Distribution",
xlab = "x", ylab = "Cumulative Probability", type = "l")
model <- lm(y ~ x)
abline(model, col = "red", lwd = 2)
```

6. Normal Distribution – qnorm()

```
x <- seq(0, 1, by = 0.02)
y <- qnorm(x, mean = 2, sd = 1)
plot(x, y, main = "Quantile Normal Distribution",
xlab = "Probability", ylab = "Quantile", type = "l")
model <- lm(y ~ x)
abline(model, col = "red", lwd = 2)
```

8. Normal Distribution – rnorm()

```
y <- rnorm(50)
hist(y, main = "rnorm", xlab = "Values", col = "blue")
```

10. Binomial Distribution – dbinom()

```
x <- seq(0, 50, by = 1)
y <- dbinom(x, size = 50, prob = 0.5)
plot(x, y, main = "Binomial Distribution",
xlab = "Number of Successes", ylab = "Probability",
type = "h")
```

12. Binomial Distribution – pbinom()

```
x <- 0:51
y <- pbinom(x, size = 51, prob = 0.5)
plot(x, y, main = "Cumulative Binomial Distribution",
```

```
xlab = "Number of Successes", ylab = "Cumulative Probability",
type = "s")
```

14. Binomial Distribution – qbinom()

```
x <- qbinom(8, size = 150, prob = 0.4)
print(x)
plot(x)
```

rbinom:

```
x <- qbinom(0.25, size = 51, prob = 0.5)
print(x)
plot(x, main = "Binomial Quantile")
```

week8

```
# Mean
x <- c(10, 20, 30, 40, 50)
result <- mean(x)
print(result)
# Median
x <- c(10, 20, 30, 40, 50)
result <- median(x)
print(result)
# Mode (user-defined function)
getmode <- function(a) {
uniq_a <- unique(a)
uniq_a[which.max(tabulate(match(a, uniq_a)))]
}
a <- c(10, 20, 10, 30, 20, 10, 40)
```

```
result <- getmode(a)
print(result)
# Covariance
x <- c(10, 20, 30, 40)
y <- c(50, 60, 70, 80)
result <- cov(x, y)
print(result)
boxplot(x, y, names = c("x", "y"), main = "Boxplot of x and y")
# Linear regression
x <- c(10, 20, 30, 40)
y <- c(50, 60, 70, 80)
result <- lm(y ~ x)
print(result)
plot(x, y, main = "Linear Regression", xlab = "x", ylab = "y", pch = 20)
abline(result, col = "red", lwd = 2)
```

week 9

2. Linear Regression

```
stdhrs <- c(1, 2, 3, 4, 5)
marks <- c(30, 40, 50, 60, 70)
student <- data.frame(stdhrs, marks)
print(student)
model <- lm(marks ~ stdhrs, data = student)
print(summary(model))
plot(stdhrs, marks,
main = "Linear Regression",
xlab = "Study Hours", ylab = "Marks",
```

```
pch = 20, col = "blue")
abline(model, col = "red", lwd = 2)
# Predict marks for the given study hours
new_student <- data.frame(stdhrs = c(6, 7))
print(predict(model, newdata = new_student))
```

3. Multiple Linear Regression

```
experience <- c(1, 2, 3, 4, 5)
qualification <- factor(c("SSC", "BTech", "MTech", "PhD", "MPhil"))
salary <- c(1, 2, 3, 4, 5)
employee <- data.frame(experience, qualification, salary)
print(employee)
model <- lm(salary ~ experience + qualification, data = employee)
print(summary(model))
plot(employee$experience, employee$salary,
main = "Experience vs Salary",
xlab = "Experience (years)", ylab = "Salary",
pch = 20, col = "purple")
abline(lm(salary ~ experience, data = employee), col = "red", lwd = 2)
```

4. Logistic Regression

```
students <- data.frame(
studyhours = c(1, 2, 3, 4, 5, 6, 7, 8),
pass = c(0, 0, 0, 1, 1, 1, 1, 1)
)
model <- glm(pass ~ studyhours, data = students, family = binomial())
print(summary(model))
students$probability <- predict(model, type = "response")
```

```
students$prediction <- ifelse(students$probability >= 0.5, 1, 0)
print(students)
plot(students$studyhours, students$probability,
main = "Logistic Regression",
xlab = "Study Hours", ylab = "Probability of Passing",
pch = 19)
lines(students$studyhours, students$probability, lwd = 2)
```

5. Poisson Regression

```
hrs <- c(1, 2, 3, 4, 5, 6, 7, 8)
ques <- c(2, 3, 4, 6, 8, 10, 13, 16)
data <- data.frame(hrs, ques)
print(data)
model <- glm(ques ~ hrs, family = poisson(), data = data)
print(summary(model))
predicted <- predict(model, type = "response")
print(predicted)
plot(data$hrs, data$ques,
pch = 19, xlab = "Study Hours",
ylab = "Questions Answered",
main = "Poisson Regression")
lines(data$hrs, predicted, lwd = 2)
```

week10

Time Series Analysis

```
sales <- c(100, 200, 300, 400, 500, 600, 700, 800, 900, 1000)
sales_ts <- ts(sales, frequency = 12)
print(sales_ts)
plot(sales_ts, type = "o",
main = "Time Series Analysis",
xlab = "Months", ylab = "Sales", lwd = 2)
```

7. Nonlinear Least-Squares Regression

```
year <- c(1, 2, 3, 4, 5, 6)
population <- c(100, 110, 125, 140, 160, 185)
model <- nls(population ~ a * exp(b * year),
start = list(a = 90, b = 0.1))
print(summary(model))
predicted <- predict(model)
print(predicted)
plot(year, population, pch = 19,
main = "Population Growth (Nonlinear Regression)",
xlab = "Year", ylab = "Population (thousands)")
lines(year, predicted, lwd = 2)
new_year <- data.frame(year = 8)
predicted_8 <- predict(model, newdata = new_year)
print(predicted_8)
```

8. Decision Tree Modeling (Regression Tree)

```
if(!require(rpart) install.packages("rpart")
if(!require(rpart.plot) install.packages("rpart.plot")
library(rpart)
```

```
library(rpart.plot)
data(mtcars)
reg_tree <- rpart(mpg ~ hp + wt,
method = "anova",
data = mtcars)
rpart.plot(reg_tree, type = 2, digits = 3,
fallen.leaves = TRUE,
main = "Regression Decision Tree (Predicting MPG)")
```

week12\

Chi-Square Test:

```
library(MASS)
data(Cars93)
str(Cars93)
car = data.frame(Cars93$AirBags, Cars93$Type)
car = table(Cars93$AirBags, Cars93$Type)
print(car)
mosaicplot(car, main="AirBags vs Car Type",
        col=rainbow(3))
print(chisq.test(car))
```

T-TEST:

```
x = c(0.593, 0.142, 0.329, 0.691, 0.231, 0.793, 0.392, 0.418)
result = t.test(x, alternative="greater", mu=0.3)
print(result)
hist(x, probability=TRUE,
```

```
    main="One-Sample t-Test",
    xlab="Sample Values",
    ylab="Density",
    col="lightblue",
    border="green")
```

F-TEST

```
x = c(66, 67, 75, 76, 82, 84, 88, 90, 92)
y = c(64, 66, 74, 78, 82, 85, 87, 92, 93, 95, 97)
result = var.test(x, y)
print(result)
boxplot(x, y, names=c("Group X", "Group Y"),
      main="Comparison of Variances",
      xlab="Groups",
      ylab="Values",
      col=c("lightblue", "lightgreen"))
```